Digital Signature Algorithm (DSA)

32. Write a C program for DSA, because the value of k is generated for each signature, even if the same message is signed twice on different occasions, the signatures will differ. This is not true of RSA signatures. Write a C program for implication of this difference?

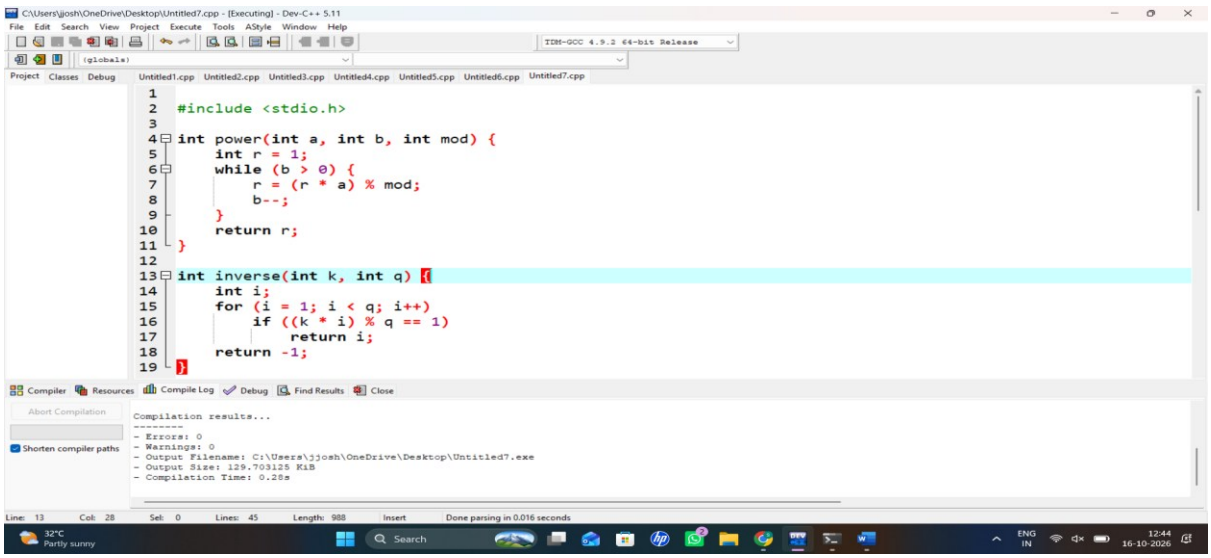
Aim:

To implement a simplified DSA signature scheme in C and demonstrate that signing the same message with different random values \(k\) can produce different signatures.

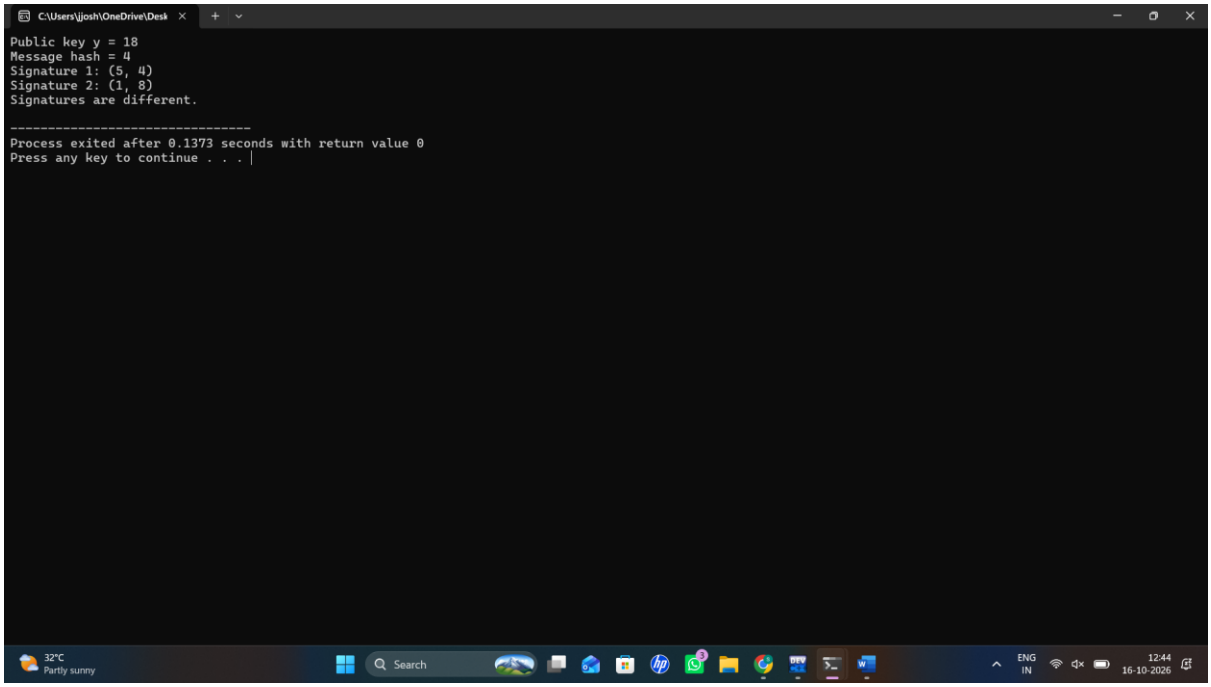
Algorithm:

1. Choose public parameters \(p=23\), \(q=11\), and \(g=4\).
2. Choose private key \(x=3\) and calculate public key \(y=g^x\bmod p\).
3. Choose a message hash \(H(m)=4\).
4. Generate a fresh secret value \(k\) for each signature.
5. Calculate \(r=(g^k\bmod p)\bmod q\).
6. Calculate \(s=k^{-1}(H(m)+xr)\bmod q\).
7. Compare the signatures generated with different values of \(k\).

Program:



Output:



Result:

The program demonstrates that DSA signatures for the same message can differ because a fresh random \(k\) is used for each signature. In contrast, textbook RSA signatures with a fixed key and deterministic padding produce the same signature for the same encoded message. Modern RSA-PSS can also use randomization, so the distinction depends on the signature scheme.